

Solution Requirements

Date	29 September 2026
Team ID	SWTID-2026-7674
Project Name	AI FAQ ASSISTANT API
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	System should allow secure login for Admin via college email ID and password with OTP verification. Students/Parents can access FAQ chatbot as guest without login, but with optional mobile OTP for personalized queries.	High
2	Authorization levels	Admin Role: Full access to add, edit, delete FAQs, view analytics dashboard, manage users. Student/Parent (Guest) Role: Only view/search answers, give feedback, cannot modify knowledge base. Super	High

		<i>Admin: Manage admin accounts.</i>	
3	External interfaces	<i>Integration with 3rd party services: 1) OpenAI GPT API for natural language understanding, 2) WhatsApp Business API for omni-channel support, 3) Google Speech-to-Text API for Tamil/English voice queries, 4) College Website Database (MySQL) to auto-fetch circulars.</i>	High
4	Transactions processing	<i>System must accept text and voice input in English and Tamil, process using NLP to understand intent, and retrieve the most relevant answer from D1 FAQ Database within 2 seconds.</i>	High
5	Reporting	<i>System should collect user feedback (Helpful / Not Helpful) and store unanswered queries in D2 Query Logs for Admin review and auto-learning of new FAQs.</i>	Medium
6	Business rules, etc.	<i>If FAQ confidence score < 70%, bot must reply "I will connect you to admin" instead of giving wrong answer. 2) Same question asked >5 times and not in DB must auto-alert admin. 3) Answers must be fetched only from verified D1 FAQ Database, not from external internet. 4) Tamil voice query must be translated to English before NLP processing.</i>	High

7	Compliance to laws or regulations	System must adhere to data privacy standards: 1) GDPR / India DPDP Act 2023: No personal data (name, phone) stored without consent, logs auto-delete after 90 days. 2) Data Security: All API keys encrypted. 3) No sharing of student data with third-party without admin approval.	High
8	Other	System should have fallback mechanism - If AI fails to answer, it should provide admin contact details and option to raise ticket. Also support multilingual input (Tamil + English) with auto language detection	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	System should process user queries quickly and provide instant answers without delay. NLP processing and database retrieval must be optimized.	Response time < 2 seconds for text query, < 4 seconds for voice query. Page/chatbot loads in < 2 seconds.
2	Scalability	System should handle growth in number of users, especially during admission season. Cloud-based architecture should scale automatically.	Supports up to 10,000 concurrent users, handles 1 lakh queries per day without crash. FAQ database scalable up to 10,000+ entries.

3	Security & Data Privacy	<i>All user queries and admin credentials must be encrypted. Prevent SQL injection, XSS attacks. API keys stored securely. No personal data leak.</i>	<i>End-to-end encryption, HTTPS only, OTP verification, 99.9% secure login. Logs auto-delete after 90 days.</i>
4	Reliability & Availability	<i>Chatbot UI should be simple, user-friendly for students, parents and even rural users. Support both Tamil and English, text and voice input. Mobile responsive.</i>	<i>90% users can get answer without training. Works on mobile, desktop, WhatsApp. Voice accuracy > 85%.</i>
5	Usability & Accessibility	<i>System should be available 24/7 with minimal downtime. Should have backup and recovery.</i>	<i>99.5% uptime (24/7 available). Downtime < 4 hours per month. Auto-backup daily.</i>
6	<i>Other</i>	<i>Maintainability: Modular code, admin can update FAQ without coding. Portability: Works on Windows/Linux/Android/iOS, deployable on AWS/Azure.</i>	<i>Bug fix <24 hrs, Deploy <30 mins, Works on all browsers</i>